DATA-DRIVEN WATER QUALITY SURVEILLANCE

TECH STACK SPECIFICATION SHEET

Project Information	Details
Document Type	Tech stack specification sheet
Week	Week 1
Prepared By	Documentation Team

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1. Project Overview

The Water Surveillance System is a web-based application designed to monitor, manage, and visualize water-related data through a centralized digital platform. The system will provide secure user authentication, real-time data management, and an intuitive dashboard for administrators and authorized stakeholders.

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2. Technology Stack

Layer	Technology	Purpose
Frontend	React 18	User Interface Development
Styling	Tailwind CSS v3	Responsive UI Design
Authentication	Firebase Authentication	Secure Login & User Management

Database	Firebase Firestore	Cloud Data Storage
Hosting	Firebase Hosting	Application Deployment
Version Control	Git & GitHub	Source Code Management
Package Manager	npm	Dependency Management
Development Environment	Visual Studio Code	Development Workspace

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3. Version Constraints

Technology	Approved Version
React	18.x
Tailwind CSS	3.x
Firebase SDK	11.x
Node.js	20.x LTS
npm	10.x
Git	Latest Stable Version

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4. System Architecture

Architecture Flow

User Interface (React + Tailwind CSS)

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Firebase Authentication

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Firebase Firestore Database

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Firebase Hosting

The application follows a cloud-native architecture where frontend services communicate directly with Firebase services for authentication, data storage, and deployment.

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5. Technology Selection Rationale

Technology	Justification
React 18	Enables fast, component-based development and efficient rendering.
Tailwind CSS v3	Provides rapid UI development with consistent design patterns.
Firebase Authentication	Ensures secure and scalable user authentication.
Firebase Firestore	Supports real-time database operations and cloud scalability.
Firebase Hosting	Offers secure, reliable, and easy deployment with HTTPS support.

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6. Domain Rationale – Water Surveillance System

The selected technology stack is suitable for a Water Surveillance System due to the following reasons:

Requirement	Solution
Secure User Access	Firebase Authentication
Centralized Data Storage	Firestore Database
Real-Time Data Availability	Firestore Real-Time Synchronization
Responsive Dashboard	React + Tailwind CSS

Fast Deployment	Firebase Hosting
Scalability	Firebase Cloud Infrastructure

Expected Benefits

- Real-time monitoring capabilities
- Secure access control
- Reduced infrastructure management
- Rapid development cycle
- Easy maintenance and scalability
- Cloud-based deployment and storage

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7. Development Standards

Coding Standards

- Use Functional React Components.
- Follow Component-Based Architecture.
- Maintain reusable UI components.
- Use Tailwind utility classes for styling.
- Follow consistent naming conventions.

Source Control Standards

- GitHub repository for project management.
- Feature branch development workflow.
- Descriptive commit messages.
- Regular code reviews before merging.

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8. Testing Strategy

Testing Area	Objective
UI Testing	Verify user interface functionality

Authentication Testing	Validate login and access controls
Database Testing	Verify CRUD operations
Integration Testing	Ensure communication between components
Deployment Testing	Validate hosted application behavior

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9. Deliverables

Deliverable ID	Deliverable
D1	React Project Setup
D2	Tailwind CSS Configuration
D3	Firebase Project Configuration
D4	Authentication Module Setup
D5	Firestore Integration
D6	Hosting Configuration
D7	Technical Documentation
